

# STAR PATTERNS

## 1. Star Patterns

1-4 square

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logic -

- Input size of square (n).
- Outer loop  $\rightarrow$  Row
- Inner loop  $\rightarrow$  Columns
- Printing star  $\rightarrow$  cout << " \* "
- Move to next line  $\rightarrow$  cout << endl

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int n;
```

```
cout << "Enter size of square: ";
```

cin 77 n;

```
for (int i=0; i<n; i++) {
```

```
for(int j=0; j<n; j++){
```

```
cout << "x=";
```

```
cout << endl;
```

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```
return 0;
```

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```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 1_Square.cpp -o 1_Square } ; if ($?) { .\1_Square }
Enter size of square: 5
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

## 2- Hollow square

```
* * * * *
```

```
*       *
```

```
* * * * *
```

```
*       *
```

```
* * * * *
```

logic-

- Input, size (n)
- Outer loop → Rows
- Inner loop → columns
- Condition for borders
  - if true → Print \*
  - else → " " space
- Line break after each row

It include <iostream>

using namespace std;

```
int main() {
```

```
    int n;
```

```
    cout << "Enter size of hollow square: ";
```

```
    cin >> n;
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < n; j++) {
```

```
            if (i == 0 || i == n-1 || j == 0 || j == n-1) {
```

```
                cout << "*" << " ";
```

```
            } else {
```

```
                cout << " " << " ";
```

```
            }
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 2_Hollow_Square.cpp -o 2_Hollow_Square } ; if ($?) { .\2_Hollow_Square }
Enter size of hollow square: 5
* * * * *
*       *
* * * * *
*       *
* * * * *
```



Logic -

- ~~row~~ Input size (n)
- outer loop  $\rightarrow$  Rows
- inner loop  $\rightarrow$  Columns
- printing start
- line break after each row

If include `<iostream>`  
using namespace std;

```
int main() {
```

cout << "Enter size of triangle: ";

০৭ ৭৭ ৭৭

```
for (int i = 1; i <= n; i++) {
```

```
for (int i = 1; i <= 1; i++) {
```

cost 22" x 11"

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cout << endl;

2

return 0;

3



Code - 1. Star\_patterns + - [ ] [ ] ... [ ] [ ]

```
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 3_Triangle.cpp -o 3_Triangle } ; if ($?) { .\3_Triangle }
```

```
Enter size of triangle: 5
```

```
PS D:\STUDY\DSA_c++\weekly_1 module\1. Star_patterns>
```

## 4. Inverted triangle

```

* * * * *
* * * *
* * *
* *
*
  
```

Logic -

- input size  $n$
- outer loop  $\rightarrow$  Rows
- Inner loop  $\rightarrow$  columns (stars per row)
- printing stars

break the break after each row

#include <iostream>

using namespace std;

int main() {

int n;

cout << "Enter size of inverted triangle: ";

cin >> n;

for (int i = n; i >= 1; i--) {

for (int j = 1; j <= i; j++) {

cout << " ";

cout << endl;

}

return 0;

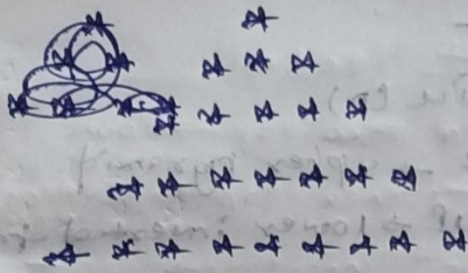
}

```

Code - 1. Star_patterns
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 4_Inverted_Tri
iangle.cpp -o 4_Inverted_Triangle } ; if ($?) { .\4_Inverted_Triangle }
Enter size of inverted triangle: 5
* * * * *
* * * *
* * *
* *
*
PS D:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns>
  
```



### 5.4 Pyramid



Logic -

- Input height (n)
- Outer loop  $\rightarrow$  Rows
- First inner loop  $\rightarrow$  ~~stars~~ before space ~~loop~~
- Second inner loop  $\rightarrow$  stars per row
- Line break after each row

It include  $\langle \text{stream} \rangle$   $\langle \text{parameters} \rangle$   $\langle \text{units} \rangle$

using namespace std; // to use std::cout

```
int main() {
```

Int  $n_2$

```
cout << "Enter height of pyramid: ";
```

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```
for (int i = 1; i <= n; i++) {
```

for  $\text{int space} = 1; \text{space} < n-1; \text{space}++ \}$

count  $< 2^n$ .

for (int j = 1; j <= 2 \* i - 1; j++) {

cout << "x";

у

```
cout << endl;
```

2)  $(-i) \leq i \leq (1-i) \text{ (für } i \geq 0 \text{)}$   
return 0;

return 0;

```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 5_Pyramid.cpp -o 5_Pyramid } ; if ($?) { .\5_Pyramid }
Enter height of pyramid: 5
    *
  ***
 *****
*****
*****

```

## 6. Diamond

```

      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *
  
```

Logic —

- Input size (n)
- First half → upper pyramid
- second half → lower inverted pyramid
- Line breaks.

```

#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter size of diamond: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int space = 1; space <= n - i; space++) {
            cout << " ";
        }
        for (int j = 1; j <= 2 * i - 1; j++) {
            cout << "* ";
        }
        cout << endl;
    }

    for (int i = n - 1; i >= 1; i--) {
        for (int space = 1; space <= n - i; space++) {
            cout << " ";
        }
        for (int j = 1; j <= 2 * i - 1; j++) {
            cout << "* ";
        }
        cout << endl;
    }

    return 0;
}
  
```

```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 6_Diamond.cpp
-o 6_Diamond } ; if ($?) { .\6_Diamond }
Enter size of diamond: 5
*
**
***
****
*****
*****
****
***
**
*
PS D:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns>
  
```



## 7.7 Half Diamond

\*

\*\*

\*\*\*

\*\*\*\*

\*\*\*\*\*

\*\*\*\*

\*\*\*

\*\*

\*

logic - 100%

• input size (n)

• first half → growing triangle

• second half → shrinking triangle

• line break - cout << endl

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter size of half diamond: ";
```

```
    cin >> n;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << " ";
```

```
        }
        cout << endl;
```

```
        for (int i = n - 1; i >= 1; i--) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                cout << " ";
```

```
            }
            cout << endl;
```

```
        }
    }
```

```
}
```

Code - 1. Star\_patterns

```
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 7_Half_Diamond.cpp -o 7_Half_Diamond } ; if ($?) { .\7_Half_Diamond }
```

```
Enter size of half diamond: 5
```

```
*
**
***
****
*****
****
***
**
*
```

```
PS D:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns>
```

logic - 100

(r) Input size ( $n$ )

- outer ~~loop~~  $\rightarrow$  Rows

- First inner loop  $\rightarrow$  leading spaces

\* Second innerloop  $\rightarrow$  start per row

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2.  $\alpha + \beta$

Q. 44 Q. 45

1-22

Pair 1st space  $\Rightarrow$  0, space  $\Rightarrow$

st. 22" 22"

$\{ \dots, i+1, i \}$

For

132 head




```
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\1. Star_patterns\" ; if ($?) { g++ 8-Half_Diamond Inverted.cpp -o 8-Half Diamond Inverted } ; if ($?) { .\8-Half Diamond Inverted }
```

```
Enter size of Half Diamond Inverted: 5
```

PS D:\STUDY\DSA\_c++\weekly\_1\_module\1. Star\_patterns>



## 2. Number Patterns

logic -

- input size ( $n$ )

- Outer loop  $\rightarrow$  Rows

- Inner loop  $\rightarrow$  column

- Printing number - cost €€ "1"

- Line break after each row.

using namespace std;

127 n°

an 77m;

```
for(int i=0; i<n; i++)
```

Count ends 1,

```

    }
    return 0;
}

```

У

Code - 2. Number\_patterns + - [ ] X

```
p -o 1_Square } ; if ($?) { .\1_Square }
```

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

1 1 1 1 1

PS D:\STUD

```
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>
```

## 2.4 Incrementing

logic :-

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

12 3 45

12 3 4 5

- Input size (n)
- outer loop  $\rightarrow$  Rows
- Inner loop  $\rightarrow$  columns (no. per row)
- printing number  $count \leq j$
- line break after each row.

It include  $\langle iostream \rangle$

using namespace std;

```
int main() {
```

int n;

```
cout << "Enter size of incrementing : ";
```

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```
for (int i=0; i<n; i++) {
```

```
for(int j=1; j<=n; j++){
```

एक लक्ष

costs  $\ll$  ends

9  
return 0;

3

```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns\" ; if ($?) { g++ 2_Incrementing.cpp -o 2_Incrementing } ; if ($?) { .\2_Incrementing }
Enter size of Incrementing: 5
12345
12345
12345
12345
12345
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>

```



## 3. Right Triangle

pattern

logic -

- Input size (n)
- Outer loop → Rows
- Inner loop → Numbers per row
- printing numbers - `cout <<`
- line break after each row

#include <iostream>

using namespace std;

int main()

{ int n;

cout << "Enter size of right triangle: ";

cin >> n;

for (int i = 1; i <= n; i++)

{ for (int j = 1; j <= i; j++)

{ cout << j;

cout << endl;

return 0;

}

Code - 2. Number\_patterns

Code - 2. Number\_patterns

```
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns\" ; if ($?) { g++ 3_Right_Tri
ngle.cpp -o 3_Right_Triangle } ; if ($?) { .\3_Right_Triangle }
Enter size of right triangle: 5
1
12
123
1234
12345
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>
```

## 4. Inverted Diamond

```

3 3 3 3 3
4 4 4 4
5 5 5
6 6 6
7

```

logic -

• Input - rows | start

• outer loop → Rows

• inner loop → leading spaces

• Number logic

int num = start + i

int count = rows - i

• second inner loop → printing number

• line break after each row.

```

#include <iostream>
using namespace std;

```

```
int main() {
```

```
int rows;
```

```
cout << "Enter number of rows: ";
```

```
cin >> rows;
```

```
int start;
```

```
for (int i = 0; i < rows; i++) {
```

```
    for (int space = 0; space < i; space++) {
```

```
        cout << " ";
```

```
    }
    int num = start + i;
```

```
    int count = rows - i;
```

```
    for (int j = 0; j < count; j++) {
```

```
        cout << num << " ";
```

```
    }
    cout << endl;
```

```
return 0;
```

```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns\" ; if ($?) { g++ 4_Inverted_Diamond.cpp -o 4_Inverted_Diamond } ; if ($?) { .\4_Inverted_Diamond }
Enter number of rows: 5
Enter starting number: 3
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>

```



## 5-4 Sandwich Pattern

### Logic-

```

3 3 3 3 3
 4 4 4
   5
  4 4 4
3 3 3 3 3
    
```

- Input starting number (start)
- Rows calculation -  $rows = (2 * start) - 1$
- middle row index -  $mid = rows / 2$
- outer loop  $\rightarrow rows$
- spaces calculation -  $spaces = (i == mid) ?$

#include <iostream>      i: row - i - 1  
using namespace std;      • number logic - num = start + spaces  
count of number per row  
• printing

```
int main() {
```

```
    int start;
```

```
    cout << "Enter starting number:";
```

```
    cin >> start;
```

```
    int rows = (2 * start) - 1;
```

```
    int mid = rows / 2;
```

```
    for (int i = 0; i < rows; i++) {
```

```
        int space = (i == mid) ? i : rows - i - 1;
```

```
        int num = start + space;
```

```
        int count = rows - 2 * space;
```

```
        for (int s = 0; s < space; s++) cout << " ";
```

```
        for (int j = 0; j < count; j++) cout << num;
```

```
        cout << endl;
```

```
    }
    return 0;
```

```
}
```

```

PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns\" ; if ($?) { g++ 5_Sandwich_
Pattern.cpp -o 5_Sandwich_Pattern } ; if ($?) { .\5_Sandwich_Pattern }
Enter starting number: 3
33333
444
5
444
33333
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>
    
```

## 6.4 Incrementing Diamond pattern.

3

44

555

6666

77777

6666

555

44

3

logic-

• Input (start/end)

• First half → Growing triangle

• Second half → Shrinking triangle

(line = 555) → on line breaks

44  
3

#include <iostream>

using namespace std;

int main()

{  
int start, end;

cout << "Enter starting number:";

cin >> start;

cout << "Enter ending number:";

cin >> end;

int up = start;

for (int i = start; i <= end; i++)

{  
for (int j = start; j <= i; j++)

{  
cout << up;

up = up + 1;

cout << endl;

int down = end - 1;

for (int i = start; i <= end - 1; i++)

{  
for (int j = end - 1; j >= i; j--)

{  
cout << down;

down = down - 1;

cout << endl;

return 0;



```
Code - 2. Number_patterns + - [ ] [ ] ... [ ] [ ] X
PS D:\STUDY\DSA_c++\weekly_1_module> cd "d:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns\" ; if ($?) { g++ 6_Incrementing_Diamond_Pattern.cpp -o 6_Incrementing_Diamond_Pattern } ; if ($?) { .\6_Incrementing_Diamond_Pattern }
Enter starting number: 3
Enter ending number: 7
3
44
555
6666
77777
6666
555
44
3
PS D:\STUDY\DSA_c++\weekly_1_module\2. Number_patterns>
```